

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** Hydrochloric acid S.G. 1.18 (~37%)

**Synonyms** Muriatic acid

**Molecular Formula** HCl  
**Molecular Weight** 36.46

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Product Code** H/1196/PB17, H/1196/PB07, H/1196/PB15, H/1196/PB08

**Address** Thermo Fisher Scientific New Zealand Ltd  
 244 Bush Road, Albany,  
 Auckland, New Zealand

**Emergency Tel.** CHEMTREC®  
**09 980 6780 or +64 9 980 6780**

**Telephone / Fax Numbers** Tel: 09 980 6700  
 Fax: 09 980 6788

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## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Substances/mixtures corrosive to metal Category 1

#### Health hazards

Acute Inhalation Toxicity - Vapors Category 2  
 Skin Corrosion/Irritation Category 1 B  
 Serious Eye Damage/Eye Irritation Category 1  
 Specific target organ toxicity - (single exposure) Category 3

#### Environmental hazards

Chronic aquatic toxicity Category 4

**Label Elements****Signal Word****Danger****Hazard Statements**

H290 - May be corrosive to metals  
H314 - Causes severe skin burns and eye damage  
H335 - May cause respiratory irritation  
H330 - Fatal if inhaled  
H413 - May cause long lasting harmful effects to aquatic life

**Precautionary Statements****Prevention**

P234 - Keep only in original packaging  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P363 - Wash contaminated clothing before reuse  
P390 - Absorb spillage to prevent material damage

**Storage**

P402 - Store in a dry place  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P406 - Store in corrosion resistant polypropylene container with a resistant inliner

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Hydrochloric acid | 7647-01-0 | 35-38    |
| Water             | 7732-18-5 | 62-65    |

## Section 4 - First Aid Measures

**Description of first aid measures****New Zealand Emergency Tel.**

CHEMTREC®  
09 980 6780 or +64 9 980 6780

**Inhalation**

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a

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|                                            |                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.                                                                                                                                                                  |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                               |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                     |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                  |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                          |

## Section 5 - Fire Fighting Measures

### **Suitable Extinguishing Media**

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Specific Hazards Arising from the Chemical**

Corrosive material. Causes burns by all exposure routes. Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Hydrogen chloride gas.

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### **Personal Precautions, Protective Equipment and Emergency Procedures**

#### **Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Do not get in eyes, on skin, or on clothing.

#### **Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

#### **Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

#### **Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

#### Advice on safe handling

Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### Conditions for Safe Storage, Including any Incompatibilities

#### Storage Conditions

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

#### Incompatible Materials

Metals. Strong oxidizing agents. Bases. sodium hypochlorite. Amines. Fluorine. Cyanides. Alkaline.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Control parameters

#### Exposure limits

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component         | New Zealand WEL                                  | Australia                                        | ACGIH TLV      | The United Kingdom                                                                                         |
|-------------------|--------------------------------------------------|--------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------|
| Hydrochloric acid | Ceiling: 5 ppm<br>Ceiling: 7.5 mg/m <sup>3</sup> | Ceiling: 5 ppm<br>Ceiling: 7.5 mg/m <sup>3</sup> | Ceiling: 2 ppm | STEL: 5 ppm 15 min<br>STEL: 8 mg/m <sup>3</sup> 15 min<br>TWA: 1 ppm 8 hr<br>TWA: 2 mg/m <sup>3</sup> 8 hr |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Individual protection measures, such as personal protective equipment

#### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material           | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                 |
|--------------------------|-------------------|-----------------|-----------------|--------------------------------------------------------------------------------|
| Butyl rubber, Nitrile    | > 480 minutes     | 0.5 mm          | AS/NZS 2161     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| rubber, Neoprene gloves. | > 480 minutes     | 0.35 mm         |                 |                                                                                |
|                          | > 480 minutes     | 0.5 mm          |                 |                                                                                |
| Viton (R)                | > 480 minutes     | 0.4 mm          |                 |                                                                                |
| PVC                      | > 480 minutes     | 0.5 mm          |                 |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Particulates filter conforming to EN 143 or Acid gases filter: Type E, Yellow. (or AUS/NZ equivalent)

**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Colorless                |                                          |
| <b>Odor</b>                                    | pungent                  |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | < 1                      |                                          |
| <b>Melting Point/Range</b>                     | -35 °C / -31 °F          |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 57 °C / 135 °F           | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | 1.8 mPa.s @ 15°C         |                                          |
| <b>Water Solubility</b>                        | Miscible                 |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Vapor Pressure</b>                          | 125 mbar @ 20 °C         |                                          |
| <b>Density / Specific Gravity</b>              | 1.18                     |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | 1.27                     | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

### Other information

**Molecular Formula** HCl

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Molecular Weight 36.46

## Section 10 - Stability and Reactivity

|                                  |                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                                         |
| Stability                        | Stable under normal conditions.                                                                    |
| Sensitivity to Mechanical Impact | No information available                                                                           |
| Sensitivity to Static Discharge  | No information available                                                                           |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                           |
| Hazardous Reactions              | Contact with metals may evolve flammable hydrogen gas.                                             |
| Conditions to Avoid              | Incompatible products, Excess heat.                                                                |
| Incompatible Materials           | Metals, Strong oxidizing agents, Bases, sodium hypochlorite, Amines, Fluorine, Cyanides, Alkaline. |

Hazardous Decomposition Products Hydrogen chloride gas.

## Section 11 - Toxicological Information

### Acute Effects

### Information on likely routes of exposure

#### Product Information

|            |                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------|
| Inhalation | Not an expected route of exposure.                                                              |
| Eyes       | Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. |
| Skin       | Avoid contact with skin. Causes burns. Skin Corrosion/Irritation.                               |
| Ingestion  | May be harmful if swallowed.                                                                    |

### Numerical measures of toxicity

#### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Based on available data, the classification criteria are not met |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Based on available data, the classification criteria are not met |

### Toxicology data for the components

| Component         | LD50 Oral               | LD50 Dermal             | LC50 Inhalation       |
|-------------------|-------------------------|-------------------------|-----------------------|
| Hydrochloric acid | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit ) | 1.68 mg/L ( Rat ) 1 h |
| Water             | -                       | -                       | -                     |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

#### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| (e) germ cell mutagenicity; | No data available                                                              |
| (f) carcinogenicity;        | No data available<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;  | No data available                                                              |
| (h) STOT-single exposure;   | Category 3                                                                     |
| Results / Target organs     | Respiratory system                                                             |
| (i) STOT-repeated exposure; | No data available                                                              |
| Target Organs               | No information available.                                                      |
| (j) aspiration hazard;      | No data available                                                              |

**Symptoms / effects, both acute and delayed**

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity** Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

| Component         | Freshwater Fish                                                            | Water Flea                 | Freshwater Algae | Microtox |
|-------------------|----------------------------------------------------------------------------|----------------------------|------------------|----------|
| Hydrochloric acid | 282 mg/L LC50 96 h<br>Gambusia affinis<br>mg/L LC50 48 h<br>Leuciscus idus | 56mg/L EC50 72h<br>Daphnia | -                | -        |

**Terrestrial ecotoxicity** There is no data for this product

### **Persistence and Degradability**

**Persistence** Persistence is unlikely, based on information available.

**Bioaccumulative Potential** Bioaccumulation is unlikely

**Mobility** Water. Soluble. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### Other adverse effects

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

**Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge.

## Section 14 - Transport Information

| Component                                | Hazchem Code |
|------------------------------------------|--------------|
| Hydrochloric acid<br>7647-01-0 ( 35-38 ) | 2RE<br>2R    |

**NZS 5433:2020**

|                      |                   |
|----------------------|-------------------|
| UN-No                | UN1789            |
| Proper Shipping Name | Hydrochloric acid |
| Hazard Class         | 8                 |
| Packing Group        | II                |

**IATA**

|                      |                   |
|----------------------|-------------------|
| UN-No                | UN1789            |
| Proper Shipping Name | Hydrochloric acid |
| Hazard Class         | 8                 |
| Packing Group        | II                |

**IMDG/IMO**

|                      |                   |
|----------------------|-------------------|
| UN-No                | UN1789            |
| Proper Shipping Name | Hydrochloric acid |
| Hazard Class         | 8                 |
| Packing Group        | II                |

**Environmental hazards** No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code** Not applicable, packaged goods

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information** None known

## Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture



**National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

| Component         | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | IMDG Marine Pollutant |
|-------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------|
| Hydrochloric acid | 25 tonne                                                                                  | 250 tonne                                                                                |                       |

**Authorisation/Restrictions according to EU REACH**

| Component         | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hydrochloric acid | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component         | CAS No    | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-------------------|-----------|-------|------|-----------|--------|-----|----------|-------|------|
| Hydrochloric acid | 7647-01-0 | X     | X    | -         | -      | -   | KE-20189 | X     | X    |
| Water             | 7732-18-5 | X     | X    | 231-791-2 | -      | -   | KE-35400 | X     | X    |

| Component         | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-------------------|-----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Hydrochloric acid | 7647-01-0 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |
| Water             | 7732-18-5 | X    | ACTIVE                                        | X   | -    | X     | -    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**Section 16 - Other Information**

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                    | <b>AICS</b> - Australian Inventory of Chemical Substances                                                                    |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>PNEC</b> - Predicted No Effect Concentration                                                                              |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail                                           |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]: Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

|                         |                |
|-------------------------|----------------|
| <b>Revision Date</b>    | 13-Mar-2023    |
| <b>Revision Summary</b> | Not applicable |

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet